



Agency for New and Renewable Energy Research and Technology

നവീനവും പുനരുപയോഗയോഗ്യവുമായ ഊർജ്ജ ഗവേഷണങ്ങൾക്കും സാങ്കേതിക വിദ്യകൾക്കുമുള്ള ഏജൻസി
(Department of Power, Government of Kerala)

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TO WHOMSOEVER IT MAY CONCERN

INTERNSHIP CERTIFICATE

This is to certify that Haritha P H, M.Sc Physics (Second Semester) student of Mar Athanasius College, Kothamangalam has successfully completed Internship Training in the topic **“Photovoltaic Technologies and Recent Innovations in Solar Panels”** under the guidance of Dr. Kamala Devi. V, Scientist F at ANERT Headquarters, Thiruvananthapuram during the period from 07/05/2025 to 15/05/2025.

Her character and conduct were found good during the internship period.



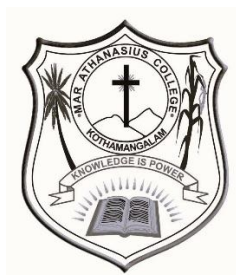
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PHOTOVOLTAIC TECHNOLOGIES AND RECENT INNOVATIONS IN SOLAR PANELS.

Project Report Submitted by

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Agency for New & Renewable Energy Research & Technology

Thiruvananthapuram, Kerala

Preface and Acknowledgement

For one week from May 7, 2025 till May 15, 2025, we did an internship at ANERT (Agency for New and Renewable Energy Research and Technology), an autonomous organization established during 1986 under societies Act by the Government of Kerala now functioning under power department; with its Head Quarters at Thiruvananthapuram.

The objective of the Agency is to gather and disseminate useful knowledge in various field of Non-Conventional Energy, Energy Conservation and Rural Technology; conduct studies, demonstrate, implement and support implementations of schemes and project in these fields and thereby deal with the problems arising out of the rapid depletion of conventional energy sources; update the technologies used In rural areas as well as introduce appropriate new technologies with an aim to reduce drudgery, increase production and improve the quality of life. ANERT is guided by an Executive Committee chaired by the Chairman, Secretary power dept; and a Governing Body chaired by the Minister of Electricity, Govt. of Kerala to provide guidelines for ANERT's activities in various energy related areas.

60

MW Solar Projects Completed in Last 10 Years

3000

MW Renewable Energy Expansion Programme

11

Awards Milestones Awards

71

MW of wind energy created since

50

Skill upgradation training programme for electricians in solar energy

30

Carbon Neutral Governance

We worked on an assignment project to enquire about the photovoltaic technologies and recent innovations in solar panels. Renewable energy is one of our areas of interest. We have always wanted to deepen our knowledge about the technologies, methods and their implementations. Through this assignment, we did not just gain a lot of theoretical knowledge but more importantly we also had the chance to sharpen our skills in a professional working environment. Not less important than the communication technologies that we have learnt is the communication skills that we have been trained and practiced through giving presentation, discussing with the supervisors, experts in the field and other staffs within the company.

We are very grateful to Dr. Kamala Devi, our supervisor at ANERT. Kamala ma'am gave us very in-time valuable instructions on various resources with which we could work on. We would also like to express our gratitude to Dr Smitha Thankachan, head of the department at Mar Athanasius College Kothamangalam for providing this internship for us.

Throughout this internship, we have also learnt many things about various renewable energies and more about photovoltaic technologies and recent innovations in solar panels, whose benefits are far beyond what we could learn in a normal project. we would like to thank ANERT HRM section for introducing us to this great opportunity in which we have developed ourselves both academically, professionally and socially.

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Chapter 1: Introduction

Basics of Solar energy



Solar energy is the most abundant energy resource on Earth. Each day, it's harvested as electricity or heat, fuelling homes, businesses, and utilities with clean, emission-free power. As the world pivots towards sustainable energy solutions, solar power is crucial in shaping our global energy landscape.

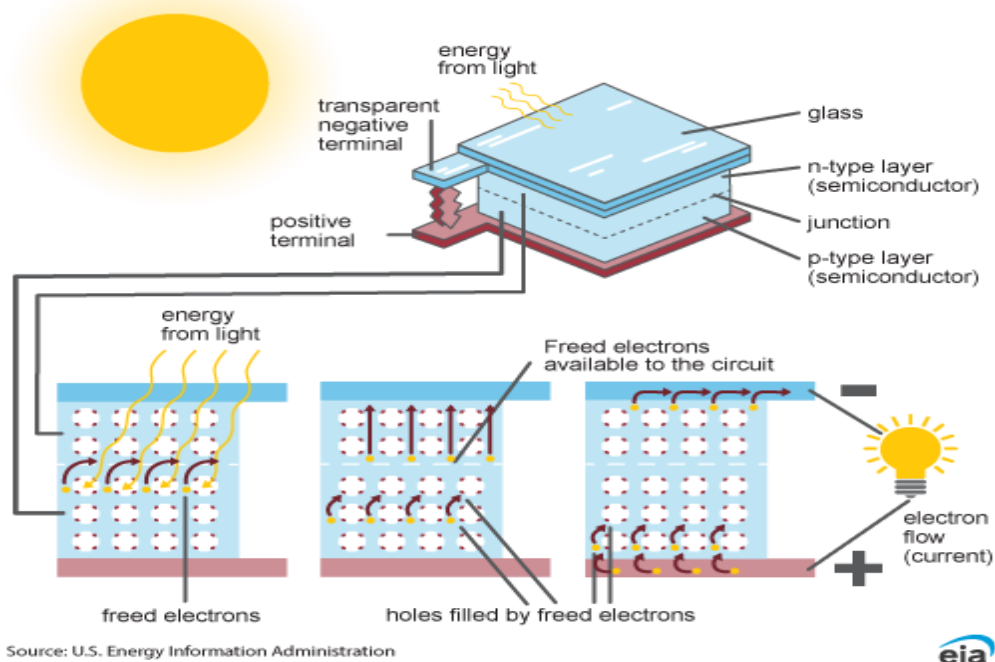
Our sun generates an infinite amount of power. Solar energy technologies capture and convert that power into electricity that we can use in our homes and businesses. Solar panels are one way to harness the power of the sun. But they aren't the only way. Solar panels, also known as photovoltaics, capture energy from sunlight, while solar thermal systems use the heat from solar radiation for heating, cooling, and large-scale electrical generation.

Our sun is basically a massive nuclear reactor. Deep in the Sun's core, nuclear fusion reactions produce huge amounts of energy that radiate outward from the sun's surface and into space in the form of light and heat.

We harness and convert solar power from the sun into usable energy using photovoltaics (more commonly known as solar panels) or solar thermal collectors.

Solar cell and its working principle

Inside a photovoltaic cell



A photovoltaic (PV) cell, commonly called a solar cell, is a nonmechanical device that converts sunlight directly into electricity. Some PV cells can convert artificial light into electricity.

Sunlight is composed of photons, or particles of solar energy. These photons contain varying amounts of energy that correspond to the different wavelengths of the solar spectrum.

A PV cell is made of semiconductor material. When photons strike a PV cell, they will reflect off the cell, pass through the cell, or be absorbed by the semiconductor material. Only the photons that are absorbed provide energy to generate electricity. When the semiconductor material absorbs enough sunlight (solar energy), electrons are dislodged from the material's atoms. Special treatment of the PV cell's surface during manufacturing makes the front surface of the cell more receptive to the dislodged, or free, electrons so that the electrons naturally migrate to the surface of the cell.

The movement of electrons, which all carry a negative charge, toward the front surface of the PV cell creates an imbalance of electrical charge between the cell's front and back surfaces. This imbalance, in turn, creates a voltage potential similar to the negative and positive terminals of a battery. Electrical conductors on the PV cell absorb the electrons. When the conductors are

connected in an electrical circuit to an external load, such as a battery, electricity flows through the circuit.

PV cells, panels, and arrays

The PV cell is the basic building block of a PV system. Individual cells can vary from 0.5 inches to about 4.0 inches across. However, one PV cell can only produce 1 or 2 Watts, which is only enough electricity for small uses, such as powering calculators or wristwatches.

PV cells are electrically connected in a packaged, weather-tight PV panel (sometimes called a module). PV panels vary in size and in the amount of electricity they can produce. Electricity-generating capacity for PV panels increases with the number of cells in the panel or in the surface area of the panel. PV panels can be connected in groups to form a PV array. A PV array can be composed of as few as two PV panels to hundreds of PV panels. The number of PV panels connected in a PV array determines the amount of electricity the array can generate.

PV cells generate direct current (DC) electricity. DC electricity can be used to charge batteries that power devices that use DC electricity. Nearly all electricity is supplied as alternating current (AC) in electricity transmission and distribution systems. Devices called inverters are used on PV panels or in PV arrays to convert the DC electricity to AC electricity.

PV cells and panels produce the most electricity when they are directly facing the sun. PV panels and arrays can use tracking systems to keep the panels facing the sun, but these systems are expensive. Most PV systems have panels in a fixed position that are usually facing directly south in the northern hemisphere—or directly north in the southern hemisphere—at an angle that optimizes the physical and economic performance of the system.

PV System Efficiency

The efficiency that PV cells convert sunlight to electricity varies by the type of semiconductor material and PV cell technology. The efficiency of commercially available PV panels averaged less than 10% in the mid-1980s, increased to around 15% by 2015, and is now approaching 25% for state-of-the-art modules. Experimental PV cells and PV cells for niche markets, such as space satellites, have achieved nearly 50% efficiency.

PV System Applications

When the sun is shining, PV systems can generate electricity to directly power devices such as water pumps or supply electric power grids. PV systems can also charge a battery to provide electricity when the sun is not shining for individual devices, single homes, or electric power grids.

Some Advantages of PV Systems Are:

PV systems can supply electricity in locations where electricity distribution systems (power lines) do not exist, and they can also supply electricity to electric power grids.

PV arrays can be installed quickly.

The environmental effects of PV systems located on buildings are minimal.

Importance of solar energy in today's energy scenario

Solar cells are crucial in today's scenarios due to their role in generating clean, renewable electricity, reducing greenhouse gas emissions, and addressing energy access challenges, particularly in remote and developing regions. They offer a sustainable solution to power homes, businesses, and even entire communities, while also contributing to a more resilient and climate-conscious future.

Objective of our internship

During our internship at **ANERT (Agency for New and Renewable Energy Research and Technology)**, we had the opportunity to explore various PV technologies ranging from traditional crystalline silicon cells to advanced bifacial and heterojunction technologies. This report aims to document our observations and understanding of the different generations of PV technologies, modern advancements, and future prospects in solar cell development.

Chapter 2: Photovoltaic (PV) Technologies

Solar cells are more complex than many people think, and it is not common knowledge that there are various different types of cells.

Classification of PV technologies;

First Generation:

Crystalline Silicon

Monocrystalline cells	Polycrystalline cells
-----------------------	-----------------------

Monocrystalline cells

Monocrystalline solar cells are made from single crystalline silicon. They are very distinctive in their appearance as they are often coloured, and the cells hold a cylindrical shape. In order to keep the costs low and performance at optimal levels, manufacturers cut out the four sides of the monocrystalline cells. This gives them their recognisable appearance.

Advantages

- They have the highest level of efficiency at 17-22%.
- They require less space compared to other types due to their high efficiency.
- Manufacturers state that this form of solar cell lasts the longest, with most giving them a 25-year warranty.
- They perform better in low levels of sunlight, making them ideal for cloudy areas.

Disadvantages

- They are the most expensive solar cells on the market, and so not in everyone's price range
- The performance levels tend to suffer from an increase in temperature. However, it is a small loss when compared to other forms of solar cell
- There is a lot of waste material when the silicon is cut during manufacture

Polycrystalline solar cells

The polycrystalline solar panels were first introduced to the public in 1981. Unlike the monocrystalline cells, polycrystalline ones do not require each of the four sides to be cut.

Instead, the silicon is melted and poured into square moulds. These then form perfectly shaped square cells.

Advantages

- The manufacturing process is cheaper and easier than the monocrystalline cells
- It avoids silicon waste
- High temperatures have fewer negative effects on efficiency compared with monocrystalline cells. This makes the polycrystalline cells more attractive to people in warmer areas as the price is lower

Disadvantage

- Efficiency is only around 15-17% due to low levels of silicon purity. So, they are not the most efficient on the market.
- They have lower output rates which make them less space efficient. So, more roof space is needed for installation.

Thin film solar cells

Thin film solar cells are manufactured by placing several thin layers of photovoltaic on top of each other to create the module. There are actually a few different types of thin film solar cell, and the way in which they differ from each other comes down to the material used for the PV layers.

The types are as follows:

- Amorphous silicon
- Cadmium telluride
- Copper indium gallium selenide
- Organic PV cells

Depending on the technology that has been used, the efficiency rates for thin film solar cells tends to vary from 7% to 13%. Since 2002, the knowledge levels and popularity for thin film solar cells have risen dramatically, which also means that research and development have been increased. Due to this, we can expect future models to hold efficiency rates of 10-16%.

Advantages

- They can be manufactured to be flexible, making them widely applicable to a range of situations and building types

- Mass production is easy to achieve, making them potentially cheaper to produce than crystalline solar cells
- Shading has a similar effect on their efficiency

Disadvantages

- They are not ideal for domestic use as they take up a lot of space
- Low space efficiency means that they will cause further expenses in the form of enhancers, like cables or support structures
- They have a shorter lifespan and so shorter warranty periods.

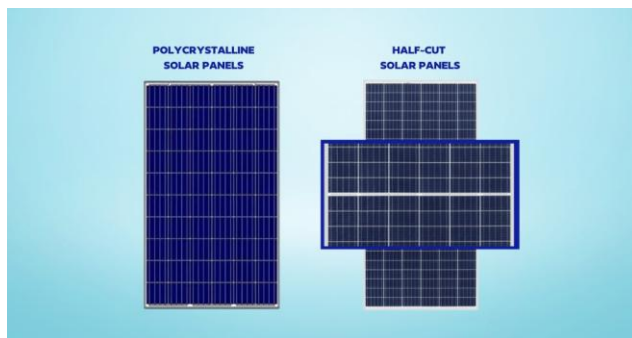
Chapter 3: Modern Innovations in PV (Photovoltaic) Technologies

Modern innovations in PV (photovoltaic) technologies focus on improving efficiency, reducing costs, and expanding applications of solar energy. Key areas of innovation include;

- Half cut
- PERC (Passivated Emitter and Rear Cell/contact)
- Bifacial Solar Panels
- HJT (Heterojunction Technology)
- Topcon (Tunnel Oxide Passivated Conduct PV)

These advancements are driving down the cost of solar energy, making it more accessible and efficient.

Half cut solar panels:



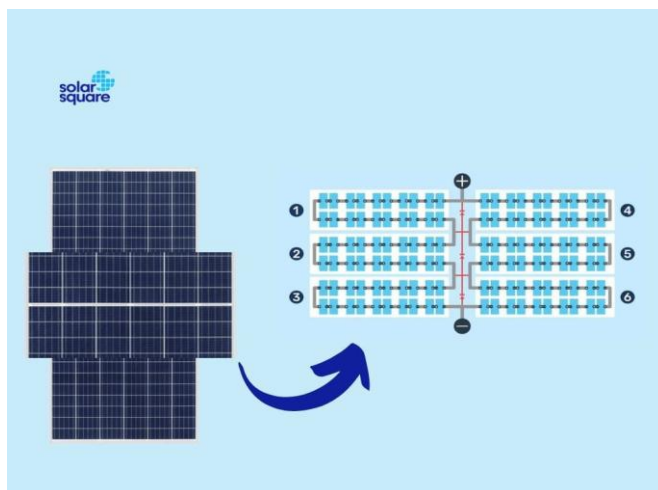
A half-cut solar panel is a modern-day technology that helps in enhancing solar power energy. These panels decrease the cell size to accommodate more cells in the system.

This technology has an improved design and consists of an anti-reflective coating or anti-reflective glass, printed silver paste

(front contact), back surface field, a doped semiconductor, and printed aluminium paste (rear connection).

These components make the Half-cut solar panels more reliable and efficient than conventional models.

How Do Half-Cut Solar Panels Work?



A half-cut solar panel works more effectively than traditional solar panel systems. In previous models, the solar panels contained 0.5V solar cells. These were combined and placed in a series. Then when the voltage was added, the cells operated at 30 V.

But in half-cut solar panels, the solar cells are cut using a laser cutter and then

added to the series. Although this process reduces the power output, the voltage stays unaffected.

Moreover, the voltage power doubles when new half-cut cells are included in the series. Hence, compared to conventional solar cells, these offer enhanced electricity and better functionality for a long time.

The Efficiency of Half-Cut Solar Panels

Half-cut solar panels are far more efficient and operate better than conventional ones. They have 120 to 144 cells that are almost double the cells in the monocrystalline solar panels.

These solar panels are made from the passivated emitter and rear cell (PERC) technology and provide high module efficiency.

Besides, in half-cut solar panels, no heavy resistive losses occur. Since the cells are cut into halves, the energy transmission from the wire to the household appliances becomes slow. Therefore, the power loss is much slower.

Additionally, the half-cut solar panels have high shade tolerance, unlike the conventional models.

In traditional models, the solar panel system stopped working if a row was shaded. But since half-cut solar panels have a different wiring arrangement, the power production does not stop if one row is shaded. All these features make half-cut solar panels better in performance than conventional solar panels.

Advantages of buying a half-cut solar panel

- There are fewer chances of power loss.
- It offers high shade tolerance.
- The half-cut solar panels provide increased low-light performance.
- The cells are compact and less prone to cracks.
- It features the latest solar design.
- It minimises the temperature of hot spots.
- It provides low electrical resistance.
- It is made of high-quality, durable materials.
- It has high longevity.

Disadvantages of buying a half-cut solar panel:

- It is expensive.
- There is a possibility of soldering.
- The cells can cause internal malfunctioning after cutting.

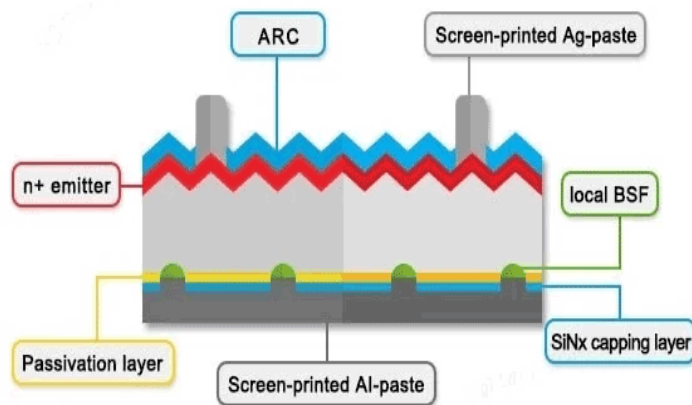
The half-cut solar panels have improved the overall output of solar energy. Though challenging to manufacture, this innovative technology provides better performance for an extended period.

These solar panel systems adhere to modern-day requirements and are more efficient than traditional solar panel versions. Homeowners can purchase this device, install it on the roof, and save electricity costs to a great extent.

PERC (Passivated Emitter and Rear Contact solar technology)

On top of the surface recombination, traditional crystalline silicon solar panels can produce losses due to inefficiency in capturing light, reflection by the solar cell itself, and partial shading by materials in the module. To reduce efficiency losses, researchers in the solar industry developed the PERC solar cell.

PERC technology was first described in the University of New South Wales in 1983 but officially registered in a paper in 1989. The PERC solar cell finally allowed PV modules to move over the 20% conversion efficiency that has been the norm for several years in the industry.



This highly efficient and improved version of c-Si technology results in PERC solar panels having a 0.86% or more increment in the efficiency of the solar cell. This provides several perks like a reduced installation time, fewer space requirements, and cost reductions by requiring fewer wires, connectors, racks, and other components that you

would require when installing the same PV capacity with traditional technology.

Pros

- Up to 1% more efficiency than traditional c-Si solar panels.
- Reduced heating absorption, allowing PV systems to perform better at high temperatures.
- Operation on a wider light spectrum.
- Better usage of the available space.
- Cost-effective technology.

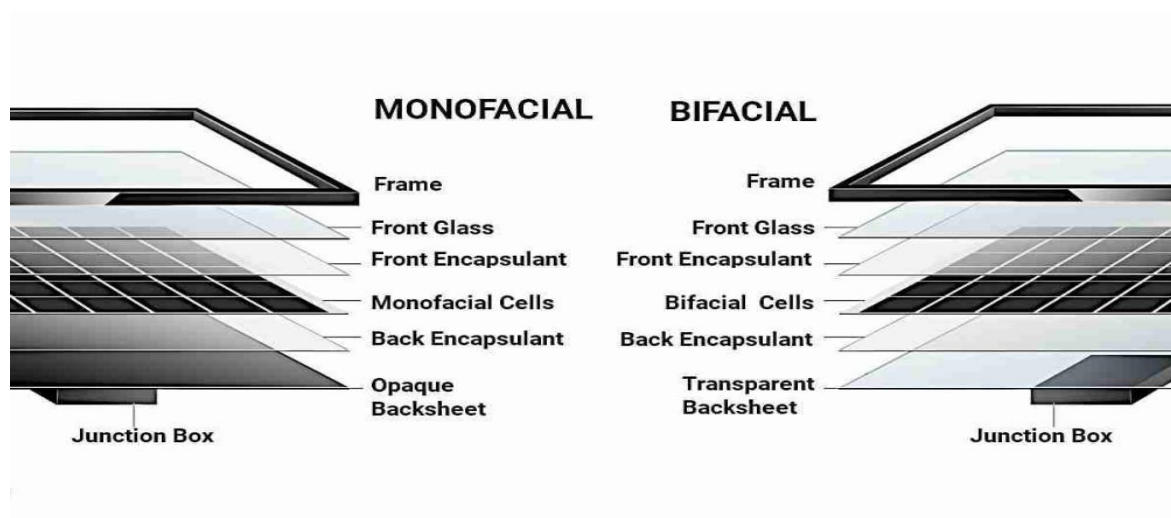
- Prices for PERC solar panels will get reduced as the technology becomes more popular.
- PERC solar cells can be combined with IBC or Bifacial technology.

Cons

- Relatively larger cost than traditional technologies.
- Light-Induced Degradation (LID) and Potential-Induced Degradation (PID) were a problem in the past for the technology. (Not anymore, since most manufacturers overcame these barriers).

The use of this technology is quite convenient for the solar industry in general. With the rise of PERC solar panels, homeowners and utilities can benefit by using less space, fewer installation components, and using PV modules with higher performance and an infinity of applications.

Bifacial Solar Panels



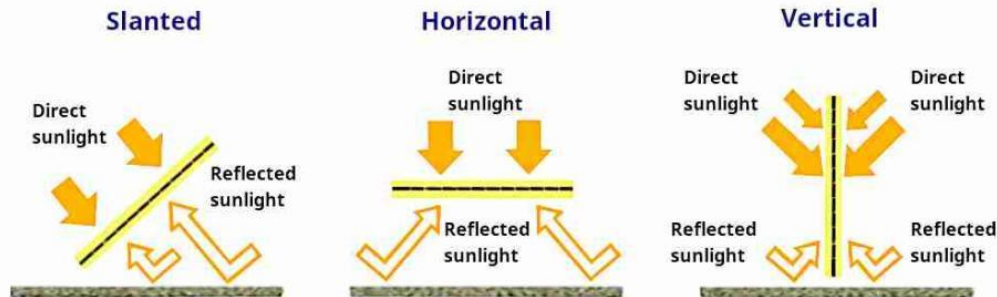
Bifacial solar panels, as the name suggests, have cells on both the front and rear sides of the panel.

This dual-sided exposure to light offers advantages in terms of total energy generation, making them suitable to utility-scale and commercial installations or buildings with flat roofs.

In terms of traditional residential installations, it has been noted that depending on how they are done, insufficient light absorption from the back of the panels might minimise the additional benefits that bifacial panels can offer.

Bifacial PV – More energy production for the same area

Several Orientation Possibilities

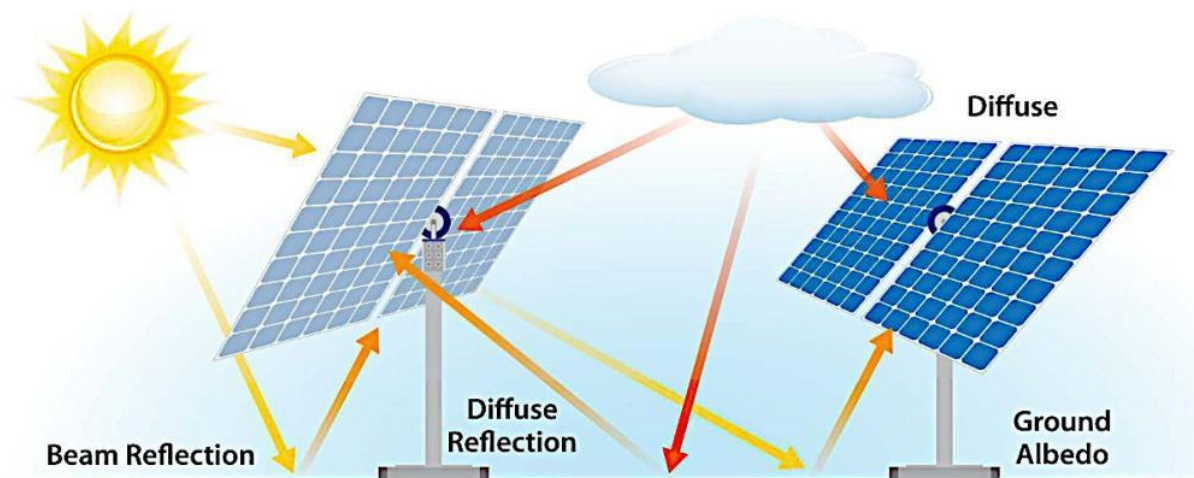


The front of a bifacial solar module is covered with a protective glass and the rear side may be made of either glass or transparent polymer back sheet that allows sunlight to pass through. This stands in contrast to conventional solar panels which have opaque back sheets.

These days, many bifacial panel designs incorporate double/dual glass at the rear of the modules. Glass-glass panels seem to better transmit light and are more resistant to unpredictable weather, moisture, corrosion, and have good mechanical load capacity.

The top solar cells of a bifacial panel capture light directly like a conventional solar panel. The bottom cells absorb light that is reflected off the ground, generating more power per square meter.

The sunlight reflected off the ground which is not absorbed is called albedo light. The ability to convert the albedo light can vary due to different factors, which in turn can affect the total panel efficiency.



- A white surface reflects most of the light that hits it and has a high albedo, while a dark surface has a low albedo. A study (Burns and McDonnell, 2019) shows that a white surface reflects more than 80% of albedo light.

- The higher the tilt angle of the module, the more light will reach the rear side, and the more energy it can produce. Panels arranged vertically are more effective, as they will have a double peak each day.

Other relevant factors affecting bifacial gains include shading, module height, orientation of the PV panels, and the distance between module arrays.

There are a number of advantages for bifacial panels

- Studies have shown that due to their ability to capture solar energy from both sides, bifacial panels can produce 10-20% more power than monofacial panels under the right site conditions. When single axis trackers are used, the additional power can be as high as 30-40%.
- Improved performance in terms of the levelized cost of energy (LCOE).
- These panels are particularly effective in areas with high albedo, such as snowy regions or white rooftops. Their ability to generate more power in a given space makes them ideal for space-constrained installations.
- Bifacial panels are typically more resistant to adverse weather conditions because of their construction. This can be confirmed by the fact that they come with longer warranties compared to traditional panels.
- Aesthetics: The transparency of the panels makes them aesthetically more appealing.

Bifacial have a few drawbacks

However, these obstacles are being slowly overcome.

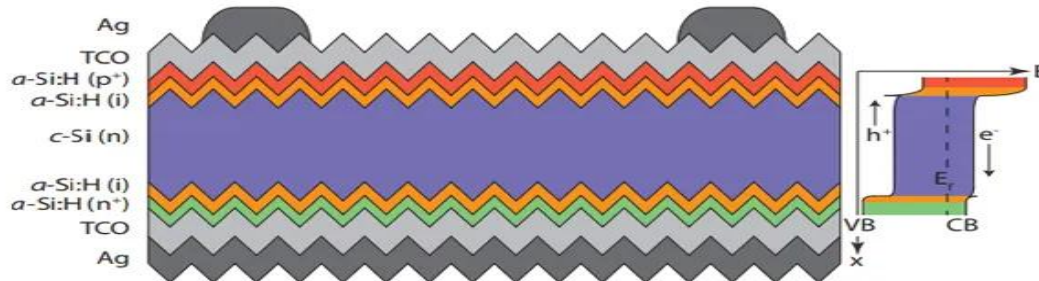
- The extra power which comes from the rear side can be inconsistent, depending on the environment and surrounding surfaces.
- The panels may require greater distance from the ground to maximise light capture.
- The cost gap between bifacial and conventional modules is decreasing but their prices are still higher comparatively.
- Care needs to be taken while installing frameless bifacial modules as overtight clamping of panels can damage the glass.
- Bifacial arrays operate at higher DC current levels than monofacial arrays, so system designs should take this into consideration.

HJT (Heterojunction Technology)

Heterojunction (HJT) solar panels were invented in the 1980s by the Japanese company Sanyo Electric (a subsidiary of Panasonic), with the first commercial products released in 1997. At the heart of this technology is to improve the efficiency of traditional solar cells by combining crystalline silicon (c-Si) with amorphous silicon (a-Si) thin-film layer to create a hybrid cell.

In HJT cells, the c-Si material used is typically monocrystalline silicon, which boasts exceptional light absorption efficiency. This material property enables it to convert a significant proportion of incident sunlight into electricity.

Typical Structure of a HJT Cell



Monocrystalline Silicon Substrate: Serves as the primary absorber layer where most of the light absorption and electron-hole (e-h) pair generation occur. In HJT cells, n-type silicon is frequently utilized as it provides enhanced carrier mobility and reduced LID.

Intrinsic a-Si Layers (i-a-Si:H): As shown in the figure, these undoped layers are placed on both sides of the c-Si substrate to provide surface passivation.

Doped Amorphous Silicon Layers: These layers are placed on top of/below the i-a-Si:H layers and doped to create the p-type and n-type regions. The p-doped a-Si layer acts as the emitter, forming a heterojunction that aids in charge separation, while the n-doped a-Si layer serves as a BSF, improving passivation and thus reducing recombination.

TCO Layers: Usually composed of ITO, ZnO or FTO, TCO layers are applied on the front (and sometimes the back surface) of the cell to act as conductive layers that also allow light to pass through, enhancing light absorption and reducing electrical losses.

Anti-Reflective Coating (ARC): It is applied to the surface of an HJT cell to reduce the reflection of sunlight and increase light absorption.

Encapsulation & Protective Layers: Encapsulation is typically done using ethylene-vinyl acetate (EVA) or polyvinyl butyral (PVB), with a front layer of tempered glass and a back sheet. Together, they protect the cell from environmental factors such as moisture, dust and mechanical damage.

Metal Electrodes

Panel frame

Pros of HJT Panels

High Efficiency: HJT panels are among the most efficient solar technologies available, with average products achieving conversion efficiencies above 22% and top-tier or bifacial products exceeding 25%.

High Bifacially: Bifacial HJT products exhibit a high bifacially factor of over 90%, making them ideal for projects that can take full advantage of reflected sunlight.

Reduced Degradation: Harnessing the benefits of premium n-type monocrystalline silicon wafers and their distinctive structure, HJT panels are less susceptible to light-induced degradation (LID), enhancing their long-term performance.

Temperature Resilience: These panels have a low temperature coefficient, meaning they maintain high efficiency even in high-temperature environments.

Good Performance in Low Light: HJT solar panels perform well under low-light conditions, such as during cloudy days or early mornings and late evenings, leading to higher energy yield throughout the day.

Versatility: These panels are generally thinner and lighter, making them applicable in various applications, including residential, C&I installations, as well as BIPV projects.

The Cons

Higher Initial Cost: Manufacturing HJT panels requires specialized equipment and expertise, which in turn results in higher initial cost. This might be a barrier for some consumers with a moderate budget.

Limited Availability: As a relatively newer technology, HJT panels are not as widely available as more established panel types. This could limit options for some consumers who have a plan to scale up.

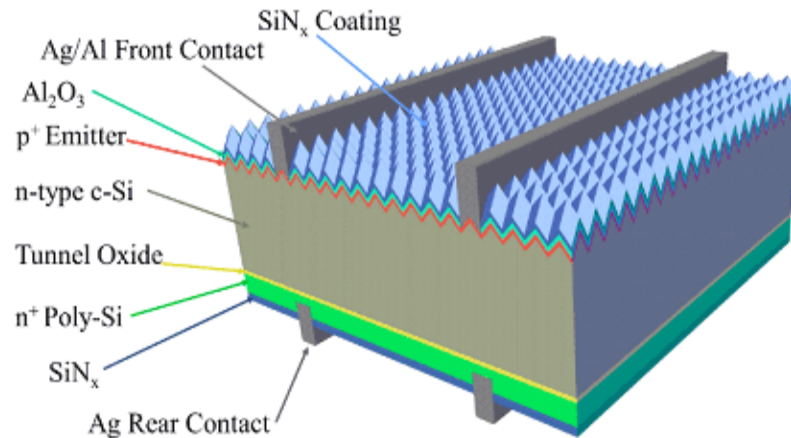
HJT solar panels improve efficiency and overall performance by transforming their semiconductor materials and associated architecture, being a landmark achievement in the industry's history.

Topcon (Tunnel Oxide Passivated Contact PV)

TOPCon solar cell technology is one of the latest trends in the solar industry, developed as an upgraded and more advanced version of the PERC/PERT technology. One of the main advantages of TOPCon solar cells is that the structure is only slightly different from PERC/PERT solar cells, meaning that manufacturers can use practically the same production line with a few improvements to produce TOPCon solar cells.

The main setback of this new technology compared to PERC/PERT is that the TOPCon solar cells require a larger amount of silver (Ag) for production. This silver increase translates into a higher manufacturing price; however, new developments might prove successful in reducing the required amount of silver, while still delivering similar or even higher efficiencies and reducing production costs.

TOPCon solar cell structure



The PERC/PERT solar cell structure has proven to be quite efficient since it was introduced to the market and implemented to manufacture PV modules, but researchers found ways to improve it and therefore created the TOPCon solar cell.

The TOPCon solar cell structure takes the base structure of the PERT solar cell but includes an ultra-thin silicon dioxide (SiO_2) layer working as the tunnel oxide layer and replaces the back surface field layer with phosphorous-doped polycrystalline silicon (n^+ Poly-Si) layer. These modifications have improved the efficiency by reducing the recombination process thanks to the passivation in the added layers.

Advantages

- Little capital investments for PERC/PERT manufacturers
- TOPCon solar cell technology is an upgrade of PERC/PERT solar cells, featuring two additional steps. This little difference between both technologies, allows manufacturers of PERC and PERT solar cells to upgrade to TOPCon by adding only a few pieces of equipment for a little investment.
- Higher efficiency reported by several manufacturers
- One of the main features of TOPCon solar panels is the higher efficiency achieved. According to the Fraunhofer ISE institute, efficiencies can go beyond 25%. This has already been proved by PV manufacturers like LONGi, Jinko Solar, and others, with a high efficiency of 26.1%.

- Better performance as a result of a lower temperature coefficient
- Traditional PV modules feature a temperature coefficient of 0.4%/°C-0.5%/°C, while PERC PV modules perform better by taking that coefficient slightly under the 0.4%/°C frame. TOPCon PV modules manufactured by Jinko Solar, on the other hand, have already proven to take the temperature coefficient to less than 0.3%/°C, highly improving their performance in many extreme weather scenarios.
-
- 15% higher bifacial factor
- The bifacial factor for PERC PV modules has been determined on average to be at around 70%. TOPCon solar panels, on the other hand, have proven to take the bifacial factor up to 85%. This increased bifacial factor can increase power gains by as much as 2%.

Disadvantages

- TOPCon solar cell technology is still facing several challenges in the manufacturing process. Some of these include different opinions on the best technique for boron deposition, different requirements for clean room conditions that have to be met, and the inability of the current selective emitter to apply to the TOPCon front emitter. For the solar industry to advance with TOPCon, these issues should be addressed.
- There is little data about mass production, limiting the roadmap for the future
- Higher silver requirement equals higher price

TOPCon solar cell technology is a revolutionizing product that upgrades the design of the PERC/PERT solar panels.

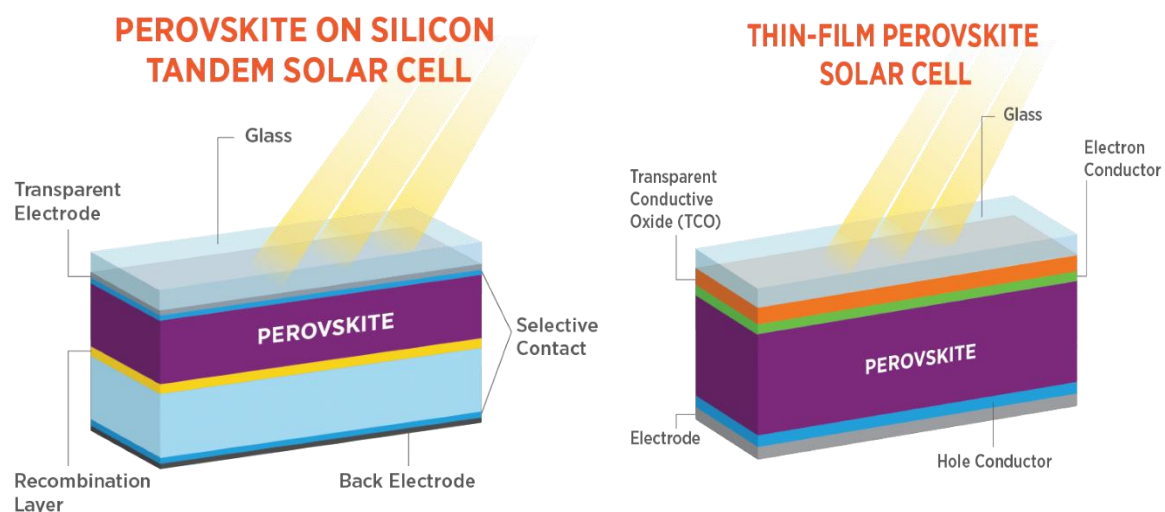
This technology has already proven its worth by reaching efficiencies that approach the Shockley-Queisser limit for single p-n junction solar cells and deliver a better performance in general.

Chapter 4: Future Advancements

Innovation in performance and manufacturing has propelled photovoltaic (PV) technology from the exception to the norm. The manifestations of innovation are defined as improvements in key technical, economic, and sustainability parameters pertaining to PV modules.

- The future cells include:
- Perovskites
- Multi-junction solar cells
- PV glass
- Floating solar farm
- AI optimized energy system

Perovskites



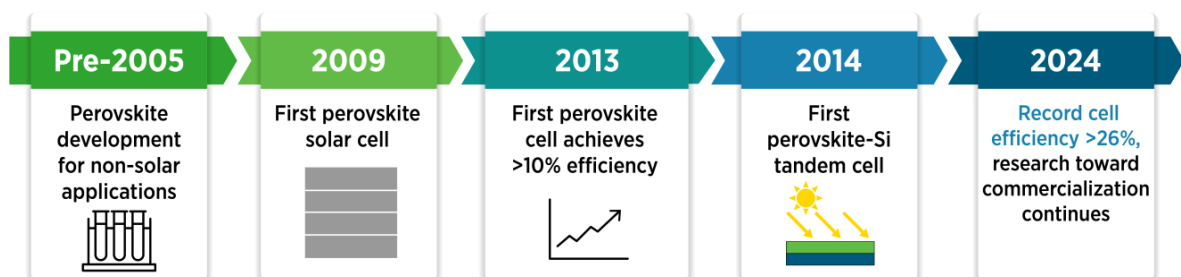
Perovskites are a family of materials that have shown potential for high performance and low production costs in solar cells. The name “perovskite” comes from their crystal structure. These materials are utilized in other energy technologies, such as fuel cells and catalysts. Perovskites commonly used in photovoltaic (PV) solar cells are more specifically called “metal-halide perovskites” since they are made of a combination of organic ions, metals, and halogens; perovskites in other applications may be made of oxygen instead of halogens and are usually entirely inorganic.

Metal-halide perovskites are the main absorbing material, or “active layer,” in a perovskite solar cell. In this potentially inexpensive technology, a thin layer of perovskite absorbs light, which excites charged particles called electrons; when these excited electrons are extracted, they generate electric power. Perovskite cells are referred to as thin-film because they require much thinner active layers relative to crystalline silicon PV.

This is possible due to perovskite's ability to absorb certain colours of light very effectively. Perovskite absorber layers can also be stacked on top of another absorber layer, such as silicon, to use the colours of light not absorbed in the perovskite, resulting in a cell that can be theoretically more efficient than cells made of either material on their own. When two layers are stacked in such a manner, this is known as a tandem solar cell.

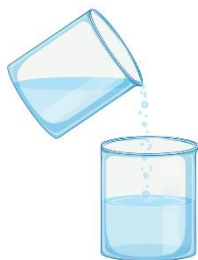
In a solar cell, the perovskite absorber is attached to other materials, which “force” electric current to flow in a single direction through the absorber layer and into the metal contacts to be collected as electric current.

Perovskite solar cells have shown remarkable progress in recent years with rapid increases in efficiency, from reports of about 3% in 2009 to over 26% today on small area devices (about 0.1 cm²). Perovskite-silicon tandem cells have reached efficiencies of almost 34%. While perovskite solar cells have become highly efficient in a very short time, perovskite PV is not yet manufactured at scale and a number of challenges must be addressed before perovskites can become a competitive commercial PV technology.



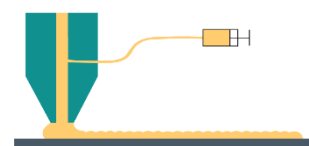
How are perovskite solar technologies made?

Because the technology is still in development, the details of each step can vary widely between research groups. This includes the elements that comprise the perovskite materials, cell and module assembly, and the tests performed on the finished product.



Material creation

Precursor salts are mixed to form a perovskite ink solution. In some cases, if the material will be deposited as a vapor, the salts are not mixed until they are deposited together on a cell in the next step.

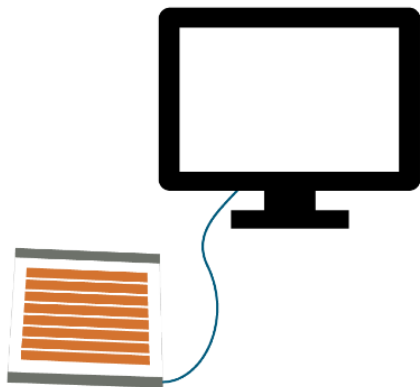


Cell production

An ultra-thin layer of perovskite ink or powder is deposited onto a base. The cell is heated to set the film. Other material layers are deposited before (and below) the perovskite layer, or after (and on top of) the perovskite layer. These layers help direct the flow of electrons.

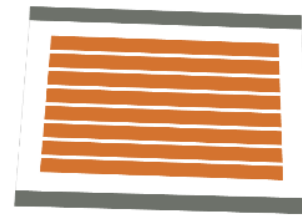
Module completion

If a module is to be created, channels are laser scribed to separate the film into individual cells. Electrical contact channels are scribed to wire those cells together. Edge sealant and encapsulant for weather proofing are added.



Testing and development

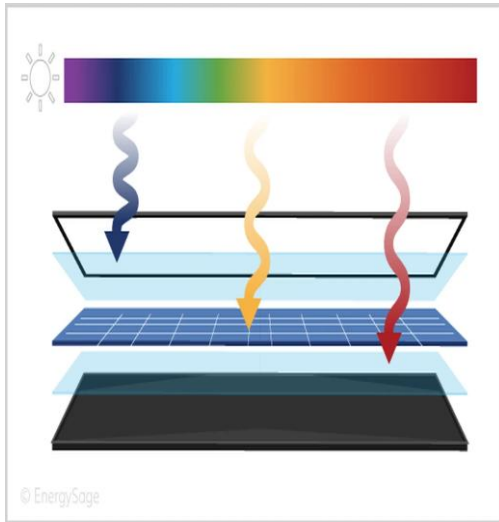
The cell or module is tested to gauge its efficiency and durability. Since perovskites are not yet in high-volume production, the knowledge gained regarding its materials and electronic properties will be used for further research and development.



Potential benefits of perovskite solar technology

- **Low-Cost Potential.** Perovskite PV cells are made using low-temperature processes and with the potential for ink-based printing of active layers. This may allow for more integrated manufacturing comprising of fewer, less expensive process steps and lower capital expenditure.
- **Use in Tandem PV Cells.** Perovskite absorber material can be stacked on top of a silicon absorber, which can convert light to electrical power more efficiently than either technology separately. Also, by changing the colour of light absorbed by the perovskite (band gap), perovskites can be engineered to pair with many other solar materials in tandem solar cells and modules.
- **Defect Tolerance.** Perovskites appear to be more resilient to imperfections in their crystal structure relative to other conventional semiconductors.
- **High Efficiency.** Perovskite material is able to convert light to electrical power at a similar efficiency to silicon.

Multi-junction solar cells

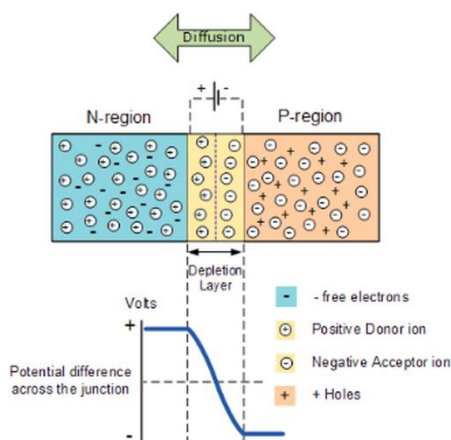


Multi-junction solar cells are capable of absorbing different wavelengths of incoming sunlight by using different layers, making them more efficient at converting sunlight into electricity than single-junction cells. While they have the potential to be many times more efficient than traditional solar cells, high production costs and continuing research and development means that multi-junction cells are not currently commercially available or feasible.

Multi-junction solar cells are a type of **tandem solar cell**, meaning they are made of stacked materials that are optimized to absorb different frequencies of

sunlight.

Multi-junction cell basis



Solar cells are made of semiconductor material, typically silicon in crystalline solar cells. Traditionally, a solar cell has two layers: an n-type with a high concentration of electrons and a p-type with a relatively low concentration of electrons. When sunlight hits the n-type layer, electrons flow from that section to the second and create an electrical current that can be captured and used for power. This type of solar cell is known as a single-junction solar cell, as it has one single boundary/junction between the n-type and p-type layers, known as a p-n junction. These p-n junctions are where electrical currents flow in solar cells.

A multi-junction solar cell is a tandem solar cell with more than one p-n junction. In practice, this means that there are multiple layers of different semiconductor materials, each of which produces electric currents in response to different wavelengths of light. This means that, theoretically, multi-junction solar cells are capable of converting more sunlight that hits them to electricity when compared to single-junction cells.

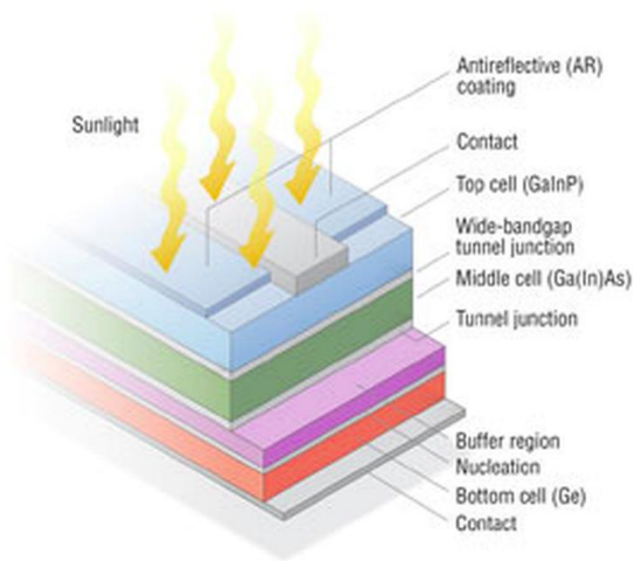
Just like normal silicon solar cells, multi-junction solar cells produce electricity through the **photovoltaic effect**. The photovoltaic effect is a complicated chemical and mechanical process, but it can be summarized in three main steps:

1. Light is absorbed by solar cells and electrons in the semiconducting material are knocked loose

2. lose electrons flow through the p-n junction between semiconductor layers, creating an electrical current
3. The current is captured and transferred to wires

Single-junction solar cells have one p-n junction to direct the flow of electricity created when sunlight hits a semiconducting material. In a multi-junction solar cell, there are multiple p-n junctions that can induce a flow of electricity.

Multi-junction solar cell structure



Multi-junction solar cells are not made using silicon as a semiconductor. Instead, materials like gallium indium phosphide (GaInP), indium gallium arsenide (InGaAs), and germanium (Ge) are used to create separate layers of semiconductors that all respond to different wavelengths of incoming sunlight.

Efficiency

A solar cell's efficiency is a measure of what percentage of incoming light that hits the cell can be converted to electricity. In terms of theoretical

efficiency, multi-junction solar cells have the potential to significantly outperform traditional single-junction solar cells. According to the Department of Energy, multi-junction solar cells with three junctions have theoretical efficiencies of over 45 percent, while single-junction cells top out at about 33.5 percent. Adding more junctions (potentially up to 5 or 6 junctions) could boost efficiency by over 70 percent. For reference, the most efficient solar panels available today have efficiencies of around 22 percent.

PV glass

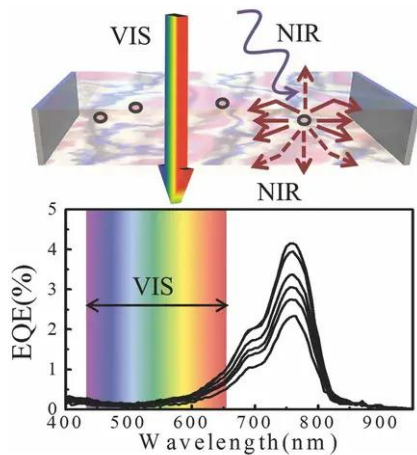
Photovoltaic glass is probably the most cutting-edge new solar panel technology that promises to be a game-changer in expanding the scope of solar. These are transparent solar panels that can literally generate electricity from windows—in offices, homes, car's sunroof, or even smartphones. Blinds are another part of a building's window that can generate electricity (we will discuss it in a later section).

Researchers at Michigan State University (MSU) originally created the first fully transparent solar concentrator in 2014. This clear solar panel could turn virtually any glass sheet or window

into a PV cell. By 2020, the researchers in the U.S. and Europe have already achieved full transparency for the solar glass.

These transparent solar panels can be easily deployed in a variety of settings, ranging from skyscrapers with large windows to a mobile device such as a phone, a laptop, or an e-reader. As these solar power windows can simply replace the traditional glass windows in offices and homes, the technology holds the potential to virtually turn every building in the United States and the world into a solar producer.

How do solar panel windows work?



A transparent solar panel is essentially a counterintuitive idea because solar cells must absorb sunlight (photons) and convert them into power (electrons). When a solar glass is transparent, the sunlight will pass through the medium and defeat the purpose of utilizing sunlight. However, this new solar panel technology is changing the way solar cells absorb light.

The cell selectively harnesses a portion of the solar spectrum that is invisible to the naked eye, while allowing the normal visible light to pass through. To achieve this technological wonder, the researchers have developed the transparent luminescent solar concentrator (TLSC) rather than trying to do the impossible by creating a transparent photovoltaic glass cell.

The TLSC is composed of organic salts that are designed to absorb specific invisible UV and infrared light wavelengths, which then glow (luminesce) as another invisible wavelength. This new wavelength is then guided to the edge of the window plastic, which thin PV solar cell strips convert it into electricity.

Once the mass production begins for transparent solar panels, researchers estimate that the TLSC should be able to deliver an efficiency of about 10%. This may not appear to be an earth-shattering number, but on a national or global scale, when almost every window in a home or office building consists of clear solar panels, the results can be transformative.

As the transparent solar panels cost comes down with their mass production and deployment, this non-intrusive technology can be scaled right from commercial and industrial applications to handheld consumer devices, while remaining very affordable.

The future of transparent solar technology

The potential to generate renewable, clean energy from the sun is enormous with transparent solar panels, considering the number of skyscrapers and buildings already in existence or under construction with a massive amount of glass surface.

Floating solar panels



Definition of floating solar (also known as floating photovoltaics or floatovoltaics /floatovoltaics): solar panel systems that float on water rather than being fixed to land or buildings.

Floating solar panels mount to structures anchored to the bed or shoreline of a water body, such as a reservoir, lake, pond, or canal. The solar array floats on top of the water, allowing the water to cool the panels, increasing their efficiency.

How do floating solar farms work?

Floating solar farms work like conventional ground-mounted farms, except they float on the water instead of being mounted on the ground. They use the same kind of solar panels and inverter equipment to generate electricity from the sunlight. The floatation devices keeping the system afloat can vary from pontoons to High-Density Polyethylene (HDPE) plastic floats.

Cables from the floating system transmit the generated electricity to connect to the grid onshore. Floating systems need extra engineering considerations compared to land-based systems, like anchoring, floatation, and water protection.

The advantages of floating solar panels?

Floating photovoltaic (FPV) systems offer several key advantages compared to conventional ground-mounted solar projects.

Improved efficiency

The cooling effect from the underlying water body lowers solar cell operating temperatures, improving power conversion efficiency. Field data shows floating PV farms consistently outperforming projected yield models with this performance boost allowing solar farmers to generate more power with water-based systems.

Dual land usage

Installing solar panels on lakes, reservoirs, or industrial ponds allows water surfaces to remain available for storage or treatment. This benefit avoids taking up land that can be used for other needs.

Reduced evaporation

Floating solar array coverage forms a protective barrier limiting sunlight exposure and wind. This shading effect decreases water loss from evaporation, saving millions of gallons annually across floating solar farms. Lower evaporation also benefits water quality.

Flood protection

When water levels rise from heavy rains or changing seasons, the floating solar panels will also rise. In this way, floating solar systems are easily able to adapt to changes in water level. Therefore, they are less likely to be damaged by flooding than solar panels fixed on land, allowing for more reliable power output.

The disadvantage of floating PV system

While floating PV offers advantages, it also comes with some drawbacks to consider:

Higher capital costs

Floating platforms, anchors, and cables add complexity and expenses over basic ground mounts. However, some cost studies show comparable or only slightly higher lifetime costs for floating systems when efficiency gains offset the initial investment.

Unknown longevity

FPV farms have operated reliably for over a decade across pilot projects, but longevity remains unproven and more data is needed into how floating solar panels hold up over decades. Things like wear and tear from weather, declining performance over time, and regular maintenance requirements must be studied over the system's full lifetime.

Grid interconnection complexity

Getting the electricity from the floating solar panels to the grid on land can be difficult as the distance the power must travel across the water can be vast. Special underwater cables, power management devices, and extra safety features are needed. These factors make connecting floating solar to the grid more complex.

What potential does floating Solar's in the green transition?

Yes, floating photovoltaic systems have the potential to generate substantial renewable solar energy globally. Some key reasons why floating PV technology promises significant growth are:

Abundant unused water areas

Lakes, reservoirs, ponds, and coastal ocean spaces offer enormous unused surface areas for floating solar installations. One estimate suggests over 400GW floating solar capacity potential across global, man-made water bodies alone.

Higher efficiency

The water's cooling effect boosts the performance of solar panels, with most floating PV systems seeing over 5% greater energy yield vs land-based equivalents. This efficiency advantage improves project cost-effectiveness.

Low environmental impact

Floating PV minimizes land usage while repurposing underutilized water surfaces. The shaded water also has ecological benefits like reduced evaporation and algae growth. To note, careful design is still required to protect waterways.

Favourable policy support

Many countries actively encourage floating solar through incentives for renewable generation. These policy tailwinds and decreasing floating PV costs support substantial industry growth projections.

Scalability and modularity

Floating solar lends itself well to starting small and then scaling up. Modular floating platform building blocks simplify expansion and customization for sites of varying sizes and shapes.

With these advantages, analysts expect over 5GW of floating solar capacity to be online by 2025 globally. So, floating PV's high scalability and efficiency give it exceptional potential for expanded renewable energy production.

AI optimized energy system

Artificial intelligence will increasingly be a differentiating factor in bringing buildings, infrastructures, and industries to their maximum energy efficiency. Concepts such as demand optimization and energy flexibility will increasingly benefit from the interconnectedness of information provided by artificial intelligence. Thanks to the use of artificial intelligence and the automatic identification of anomalies and energy saving measures, decision-makers will be able to spend their time on what is important, prioritizing those actions with the greatest impact.

Energy transformation and artificial intelligence: a promising encounter

In the last decade, we have witnessed a quiet revolution that has permeated every corner of our society: artificial intelligence (AI). From explainable algorithms to deep learning systems, AI has evolved by leaps and bounds, surpassing even human capability in complex tasks. But what happens when we couple this powerful tool with one of the most pressing challenges of our time: climate change?

Climate change is an undeniable reality. Temperatures are rising, weather patterns are altering, and natural resources are under threat. In this context, environmental policies have become a key pillar to mitigate the effects of climate change and promote a more sustainable use of energy resources. This is where energy efficiency comes into play: the idea of achieving more with less, using as little energy as possible to achieve the same results.

But can artificial intelligence contribute to this cause? The answer is loud and clear: yes, AI can be a powerful ally in the quest for greater energy efficiency in the energy sector. In this article we will look at some of the ways in which AI can make a difference in the field of energy efficiency.

Automatic detection of anomalies in energy consumption

By monitoring energy consumption, AI develops customized predictive models to predict consumption patterns based on several variables such as time of day, weather, asset type, occupancy, usage and other relevant factors. By predicting consumption patterns, AI automatically detects unusual deviations in real time.

Anomaly detection can be performed using a variety of approaches:

Asset-centric approach: AI analyses the specific behaviour of a building, infrastructure, or industry. If consumption deviates significantly from what is expected, an alert is triggered.

Comparison with similar assets: AI compares the consumption of an asset with data from other similar assets. This allows us to identify those assets with the greatest potential for improvement and prioritizing efforts.

Among the benefits of this intelligent anomaly detection:

Reduced Overconsumption: Early detection of anomalies avoids unnecessary expenses and contributes to sustainability.

Strategic Focus: Companies with multiple assets can focus their investments where it really matters. AI pinpoints those assets with the most room for improvement and where to focus efforts.

Chapter 5: Conclusion

Our internship at ANERT provided us with valuable exposure to the field of photovoltaic technology and its dynamic innovations. From conventional crystalline silicon cells to advanced structures like PERC, HJT, and TOPCon, we observed how continuous research and development have significantly improved the efficiency, performance, and application versatility of solar panels.

The integration of novel technologies such as bifacial modules and heterojunction cells into the commercial market demonstrates the industry's commitment to achieving higher energy yields and longer operational lifespans. Furthermore, emerging concepts like perovskite solar cells and tandem modules indicate a promising future for solar technology in meeting global energy demands sustainably.

In conclusion, our study highlights the importance of innovation in the photovoltaic sector and the role of organizations like ANERT in promoting renewable energy adoption and awareness. We hope this knowledge will inspire further research and application of clean energy solutions.

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